



Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Test: Tuesday 6 October — the whole of C.1

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Recognising and defining SHM

- ☐ State the defining condition: acceleration (or force) is proportional to displacement from equilibrium, and directed **towards** it — $a = -\omega^2 x$, $F = -kx$.
- ☐ Recognise SHM from an a - x or F - x graph (straight line through the origin, negative gradient), and explain why a given system does or does not oscillate with SHM.
- ☐ State the condition for the model to be valid (small angle, small displacement, or within the spring's Hooke's-law range).

2. Position, velocity and acceleration through the cycle

- ☐ State where displacement, velocity, acceleration, KE and PE are each zero or maximum — at equilibrium and at the extremes.
- ☐ Determine the direction of velocity or acceleration at a given instant, and whether the object is speeding up or slowing down.
- ☐ Work out where the oscillator is after a stated fraction of a period ($T/4$, $T/2$, $3T/4$), starting from an extreme or from equilibrium.

3. Period, frequency and ω

- ☐ Read amplitude, period and frequency off a graph, and convert between T , f and ω ($f = 1/T$, $\omega = 2\pi f$).
- ☐ Read parameters from a time equation such as $x = A \sin(\omega t)$.
- ☐ Use a **ratio**, not two full calculations, when a quantity is scaled (e.g. mass tripled, length halved).

4. Graphs of x , v , a and non-time graphs

- ☐ Sketch or translate between x - t , v - t and a - t graphs for the same motion, given the initial conditions.
- ☐ Sketch a v - x graph (an ellipse, not a sine curve) and a KE- x or PE- x graph.
- ☐ Use the **gradient** and **area** of a graph correctly: gradient of x - t is velocity; area under v - t is displacement.

5. Quantitative acceleration, springs and pendulums

- ☐ Calculate acceleration at a stated displacement using $a = -\omega^2 x$, including the direction.
- ☐ Use $T = 2\pi\sqrt{m/k}$ for a mass-spring system and $T = 2\pi\sqrt{L/g}$ for a simple pendulum, and state what changing m , k , L or g does to T .
- ☐ Obtain ω , f or T from the gradient of an a - x graph.

6. Speed, energy and experimental work

- ☐ Use $v = \pm\omega\sqrt{A^2 - x^2}$ to find speed at a stated displacement.
- ☐ Sketch and interpret KE, PE and total-energy graphs against displacement.
- ☐ Describe how to measure T (or find g or k) from practical data, and read a period correctly off an energy graph (energy repeats every $T/2$).

Remember: $a = -\omega^2 x$ $T = 2\pi\sqrt{m/k}$ $T = 2\pi\sqrt{L/g}$ $v = \pm\omega\sqrt{A^2 - x^2}$ energy repeats twice per cycle, so its period is $T/2$

Practise with the C.1 Review Packet on your class page — due Mon 5 Oct.